FACULTY OF COMPUTING

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SECI1113- SYSTEM ANALYSIS AND DESIGN

SECTION 01

PROJECT PHASE 1

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carmeliasani/FlexHub_FlexRent_System_Project1_SAD_20252026

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1.0 INTRODUCTION

In today's fast-paced and dynamic lifestyle economy, the demand for flexible options in securing event spaces, short-accommodations and specialized professional services has significantly increased. Securing venues or specialized expertise for single events through traditional, long-term contracts is costly and highly impractical. As a result, multi-vertical digital marketplaces that offer mixed time intervals and on-demand matching have become a popular alternative. These marketplaces bridge the gap by offering diverse booking intervals and automated matching.

This project proposes the development of FlexRent, an intelligent, web and mobile-based marketplace migration system designed to support, automate and improve the operations of lifestyles sectors and professional service directories. The platform allow users to browse available listings, evaluate real-time calendar availability, secure immediate bookings and manage transactions seamlessly through an integrated automated interface. By completely replacing obsolete, manual messaging workflows, this system drastically minimizes operational delays and errors for all parties involved.

The FlexHub platform integrates advanced Artificial Intelligence (AI) capabilities, such as AI Smart Match, dynamic pricing, receipt or ID scanning or schedule optimization. It is to enhance backed functionality and improve decision-making for all users. The system natively supports a dual-role management, allowing individuals to act as both renter or client and owner or expert to either book services or list assets within the same platform. Ultimately, this project aims to eliminate manual messaging process by replacing fragmented manual workflows with a highly efficient, automated and secure system.

2.0 BACKGROUND STUDY

Currently, the short term property rental, event space and professional service sectors heavily rely on fragmented, semi-digital models. In the existing landscape, listing entities utilize static online directories to advertise their assets. To rent a space or a dress, consumers must manually initiate contact with individual owners via external messaging applications like WhatsApp to cross reference scheduling availability.

This manual approach introduces severe challenges. For providers, tracking is completely unautomated. The asset owners must physically cross reference paper notebooks or personal calendars to verify hourly, daily or monthly availability. This will lead to data fragmentation and accidental double booking. For consumers. The process will causes long communication delays to check pricing and also availability. In addition, financial transactions require users to manually transfer funds and text screenshots of receipt, forcing providers to log into banking portals to verify funds before a slot is locked. This administrative burden worsens during cancellations and modifications, which require manual computation of refund penalties and manual adjustments to the schedule.

As technology continues to evolve, especially in Artificial Intelligence (AI), traditional rental services now have the chance to become more efficient and automated. An integrated, AI-powered platform will replace these human-dependent workflows with a centralized systems. An AI-powered system can help businesses work more smoothly, reduce mistakes and provide a better experience for customers.

3.0 PROBLEM STATEMENT

The current FlexHub system currently operates using a highly manual workflow that depends heavily on third-party messaging platforms such as WhatsApp for communication, booking coordination, and payment confirmation. Most interactions between users and providers are conducted through WhatsApp, an app for communication. This manual process creates operational inefficiencies, delays and a high risk of human error, affecting both users and service providers.

One of the major problems identified in the current "AS-IS" system is the lack of an automated booking management system. When users asked about available booking slots, providers need to manually check their physical notebooks or personal calendars before responding. This causes slow response time, double-booking risks and scheduling conflicts, especially when handling multiple customers simultaneously.

Besides that, payment processing is also performed manually. Providers need to send their bank account details through WhatsApp, while users must transfer the payment and send the receipt as proof. Provider then manually verify transactions through their banking portals before confirming bookings. This process is time-consuming, inefficient and vulnerable to fraud or missing payment records.

The current system also lacks an efficient booking modification feature. Users who wish to reschedule or modify their bookings must contact providers directly through chat. Providers are then required to manually update schedules and rearrange booking slots. This may lead to inaccurate booking records.

Furthermore, cancellation and refund handling are entirely manual. Providers must manually calculate refund amounts based on their own policies, collect customer banking information and manually transfer refund payments. This process can delay refund handling and may result in calculation errors or customer dissatisfaction.

Overall, the dependency on manual processes reduces operational efficiency, limits scalability and negatively impacts user experience. Therefore, an automated AI enhanced system is needed to optimize booking operations, payment validation, schedule modification processes and refund management while enhancing system accuracy, operational efficiency and overall customer satisfaction.

4.0 PROPOSE SOLUTION

To address the challenges identified in the problem statement, this project proposes the "TO-BE" system is designed to replace the current manual workflow with a fully integrated web and mobile platform that automates booking management, payment processing, customer communication, and scheduling processes. By eliminating the dependency on WhatsApp communication, the system can improve operational efficiency, reduce human errors, and provide a more organized booking experience

The proposed platform will include several automated features such as real-time booking availability, integrated online payment gateways, automated notifications, and self-service dashboards. Users will be able to check available booking slots instantly without waiting for manual responses from providers. Once a booking is confirmed, the system will automatically update schedules and send booking confirmations, payment updates, and reminder notifications to users and providers.

In addition, the payment process will be fully digitalized through a secure payment gateway integrated into the platform. Users can make payments directly through the system, while providers can receive automatic payment verification without manually checking bank transfers or payment receipts. This feature helps reduce transaction processing time and minimizes the risk of payment fraud or missing records.

The proposed system will also support automated booking modification and cancellation management. Users can reschedule or cancel bookings directly through the platform, while the system automatically updates schedules and manages slot availability. Refund calculations will also be processed automatically based on predefined cancellation policies, reducing manual workload and improving accuracy.

The proposed system will also integrate the AI Dynamic Pricing feature to make price management more efficient and flexible. In the current system, pricing is handled manually, which can lead to inconsistent prices and difficulties in adjusting prices based on customer demand. With the AI Dynamic Pricing feature, the system can automatically adjust rental or service prices in real time according to factors such as booking demand, seasonal trends, service availability, and booking frequency. This allows service providers to manage pricing more easily without needing to manually update prices all the time. At the same time, customers can receive fair and updated pricing information based on current demand. By automating the pricing process, the system can help reduce workload, improve operational efficiency, increase booking opportunities, and provide a better overall experience for both customers and service providers.

In addition, the proposed system will implement the AI Smart Match feature to provide personalized recommendations for users. The system analyzes user preferences booking history, selected categories and uploaded reference images to suggest suitable dresses or costumes automatically. This helps users find rental items more quickly and improves their overall booking experience. At the same time, providers can increase booking opportunities because their listings are matched with relevant customers. By integrating AI Smart Match, the FlexRent system becomes more efficient, user-friendly and intelligent for both customers and providers.

4.1 Technical Feasibility

The proposed system is technically feasible as it can be developed using widely available and reliable technologies. The system will be implemented as a web-based application using :

- HTML, CSS and JavaScript for the frontend
- PHP for the backend
- MySQL for the database

In addition, the system will include important features such as :

- User registration and login system
- Booking and time slot management system
- Inventory management schedule
- AI-based Smart Matching feature
- Dashboard for both seekers and providers

These technologies are commonly used, easy to maintain and suitable for developing the proposed system.

4.2 Operational Feasibility

The system is operationally feasible as it improves the current manual rental process and makes it more efficient and user-friendly. Currently, booking and inventory management are done manually, which can lead to errors such as double bookings and poor tracking of costume availability. The proposed system helps solve these problems by :

- Allowing users to browse and book costumes online
- Automatically managing booking schedules with time slot and buffer time features
- Enabling providers to manage inventory efficiently
- Offering AI-based recommendations to improve user experience

Overall, the system is designed to be simple and easy to use, requiring minimal training for users.

4.3 Economic Feasibility

From a financial perspective, the proposed system is economically feasible as it requires relatively low development costs while providing long-term benefits. The main costs are justified by the benefits such as :

- Reducing manual work and operational errors
- Saving time for both users and providers
- Improving customer satisfaction
- Increasing efficiency in managing bookings and inventory
- Supporting business growth through better system management

Based on these advantages, the proposed system provides good value and is worth developing.

Cost Benefits Analysis (CBA) - Dress and Rental Management System

1. Financial Assumptions

Item	Value
Discount rate	10%
Sensitivity Factor (cost)	1.10
Sensitivity Factor (growth)	1.15
Annual Cost Growth	8%
Annual Benefit Growth	12%

2. Development Costs (Year 0)

Item	Cost (RM)
Hardware (PC, server setup)	2500
Software (tools)	800
Development (team effort)	3200
Training & testing	1000
Total Development Cost	7500

3. Annual Operating Costs

Cost type	Year 1	Year 2	Year 3	Year 4	Year 5
Hosting & maintenance	1000	1080	1166	1259	1360
System maintenance	800	864	933	1008	1089
Admin operation cost	1500	1620	1750	1890	2041
Total Annual Cost	3300	3564	3849	4157	4490
Annual prod.Costs (Present Value)	3000	2964	2892	2840	2788
Accumulated Costs	10500	13464	16356	19196	21984

4. Benefits (Estimated Savings)

Benefit Type	Year 1	Year 2	Year 3	Year 4	Year 5
Reduced booking error (time-slot system)	1500	1680	1882	2108	2361
AI Smart Matching	3000	3360	3763	4215	4721
Time saving	2000	2240	2509	2810	3147
Total Benefits	**6500**	**7280**	**8154**	**9133**	**10229**
Annual production Benefit (Present Value)	5909	6017	6126	6238	6351
Accumulated benefits	5909	11,926	18,052	24,290	30641
Gain or Loss	(4591)	(1538)	1696	5094	8657
Profitability index	8657/7500 = 1.15				

In conclusion, the proposed Dress and Costume Rental System effectively addresses the limitations of traditional rental services by improving efficiency, reducing errors and enhancing user experience. Features such as time-slot management help prevent booking conflicts, while the AI-based Smart Matching feature simplifies costume selection.

Based on the feasibility analysis, the system is technically, operationally and economically feasible, with a Profitability index greater than 1. Therefore, the system is considered beneficial and recommended for implementation.

5.0 OBJECTIVES

Objective of our FlexRent include :

1) To analyze the current manual booking and service management process used in FlexRent, including booking slots, checking payment, booking modification and cancellations through WhatsApp.
2) To identify the weakness and inefficiencies of the existing manual system such as delayed responses, double bookings, payment verification issues and poor record management.
3) To design an automated and integrated FlexRent platform that improves booking management, payment processing, scheduling and customer communication.
4) To develop functional requirements for the proposed FlexRent system to ensure all booking, payment, modification and cancellation processes can be operated efficiently and systematically.
5) To determine the nonfunctional requirements of FlexRent including system security, reliability, usability, performance efficiency and data accuracy.
6) To integrate Artificial Intelligence(AI) features into FlexRent in order to provide smarter, faster and more personalized services for users and providers.
7) To improve user experience by providing real time availability checking, automated payment verification, self service booking management and instant notifications.
8) To create a scalable and efficient marketplace platform that supports both service providers and customers through digital automation and management.

6.0 SCOPE OF THE PROJECT

We are developing a system that helps users manage dress and rental more efficiently through a centralized platform.The scope of this project focuses on the renting sector (FlexRent), specifically managing physical rental items across flexible time unit. The system allows users to register an account using their email address and securely log in to access to the system. This platform is designed to improve the rental process by providing features such as booking management, inventory tracking and intelligent recommendations.

In addition, the system support two main types of users which are “Seekers” and “Providers”. Each type of user will have different function and system based on their roles. However, users are allowed to switch roles within the same system, proving more flexibility and usability.

For Providers (owner) :
In this system, providers are allowed to manage their costume listings efficiently. They can add, update and remove dress or costumes, as well as monitor the availability and condition of each item through the inventory management feature. Through the AI dynamic Pricing, the system automatically optimizes the providers hourly and daily rental rates in real-time based on local demand peaks, event specific demands, and platform scarcity to maximize their rental revenue. The system's AI Smart Match also directly impacts providers by automatically channeling and recommending their specific costume listings to the most relevant seekers, go to the higher conversion rates based on style matching and past trends. The system also helps providers manage bookings by automatically organizing schedules and preventing overlapping reservations using time slot management system. Providers can track all incoming bookings and ensure that costumes are properly prepared between rental periods.

For Seekers (customer) :
In this system, seekers are able to browse available dresses and costumes easily based on categories, sizes and preferences. They can customize their renting duration based on their specific needs, choosing to rent items by the hour.When browsing, seekers will experience real-time flexible rates driven by AI Dynamic Pricing, allowing them to book during off-peak hours for better deals. The system also provides an AI Smart Match featuring the user's profile, past bookings, and reference images to automatically suggest highly tailored listings that match their style. to analyze Seekers can check costume availability in real time and bookings through the system. The system ensures that all bookings follow proper time slot management, where conflicts are avoided and buffer time is included between rentals for cleaning and preparation. In addition, seekers can view their booking history and modify or cancel their reservations when necessary.

To deliver this renting system successfully, the total estimated time required to analyze, design, and complete the interactive prototype in Figma is 14 weeks. This duration covers everything from gathering user requirements, designing the data flow and renting system logic, up to completing the final simulation.

7.0 PROJECT PLANNING

Project planning outlines the timeline, phases and activities required to successfully develop the Dress and Costume Rental System. It ensures that all tasks are organized, resources are allocated efficiently and milestones are achieved within the given timeframe. The project will be divided into several key phases.

1. Phase 1: Requirement Analysis and System Design
    - Collect and analyze user requirements, identify system functions and prepare specifications.
    - Design the system architecture, database schema and user interface prototypes

   Expected Output: System Requirement Specification (SRS), design diagrams and UI mockups.

2. Phase 2: Development and Integration
    - Implement system modules including booking management, inventory tracking and AI Smart Matching.
    - Integrate database and ensure smooth interaction between Seekers and Providers.

   Expected Output: Functional prototype of the system.

3. Phase 3: Testing, Documentation and Deployment
    - Conduct unit testing, integration testing and user acceptance testing.
    - Prepare technical documentation, user manuals and final project report.
    - Deploy the system for use.

   Expected Output: Tested system, documentation and final presentation.

The project is expected to be completed within 14 weeks, with each phase allocated sufficient time to ensure quality and reliability.

7.1 HUMAN RESOURCES

The project team consists of five members, each assigned a distinct role to ensure balanced workload and effective collaboration.

1. Project Manager
   - Responsible for overall project planning, coordination and supervision.
   - Overseas project progress and ensures all tasks are completed within the timeline.
   - Manages communication among team members and resolves any project-related issues.
   - Ensures that project objectives and deliverables meet the required standards.
   - Prepares progress reports and presents updates to supervisors or evaluators.

2. System Analyst
   - Conducts requirement analysis and gathers user needs through observation and research.
   - Defines system specifications and functional requirements clearly.
   - Ensures that the proposed design aligns with both user and business needs.
   - Collaborates with the designer and developer to maintain consistency between analysis and implementation.
   - Documents all findings and system requirements for future reference.

3. System Designer
   - Develops the overall system architecture and database structure.
   - Designs user interface layouts that emphasize usability and efficiency.
   - Creates prototypes and wireframes to visualize system flow.
   - Ensures that design components support scalability and maintainability.
   - Works closely with the analyst and developer to translate design into implementation.

4. Programmer / Developer
   - Implements system modules based on design specifications.
   - Integrates AI features such as Smart Matching.
   - Performs coding, debugging and optimization to ensure system performance.
   - Collaborates with the designer to ensure proper functionality of the user interface.
   - Maintains version control and documentation of source code.

5. Tester and Documentation Specialist
   - Conducts unit testing, integration testing and user acceptance testing.
   - Identifies and reports system errors or inconsistencies for correction.

- Ensures that all modules function correctly and meet project requirements.
- Prepares technical documentation, user manuals and final project reports.
- Assists in preparing presentation materials for project evaluation.

7.2 WBS

The Work Breakdown Structure (WBS) is used to break down the development of the Dress and Costume Rental System into manageable phases and tasks. This helps ensure that all activities are well-organized, clearly defined, and effectively executed throughout the project.

1.0 Project Planning

- 1.1 Define project objectives
- 1.2 Determine project scope
- 1.3 Prepare project proposal
- 1.4 Assign team roles

2.0 Requirement and Analysis

- 2.1 Conduct stakeholder interview
- 2.2 Gather system requirements
- 2.3 Analyze current rental process
- 2.4 Identify problems
- 2.5 Define non-functional/functional requirements
- 2.6 Create user flow

3.0 System Design

- 3.1 Design system architecture
- 3.2 Design user interface
- 3.3 Design database
- 3.4 Design Data Flow Diagram
- 3.5 Define booking workflow

4.0 System Development

- 4.1 User management
- 4.2 Product catalog and search
- 4.3 Booking system
- 4.4 Inventory management
- 4.5 Payment system
- 4.6 AI feature

5.0 Testing and Debugging

- 5.1 Perform system testing
- 5.2 Identify errors
- 5.3 Fix bugs

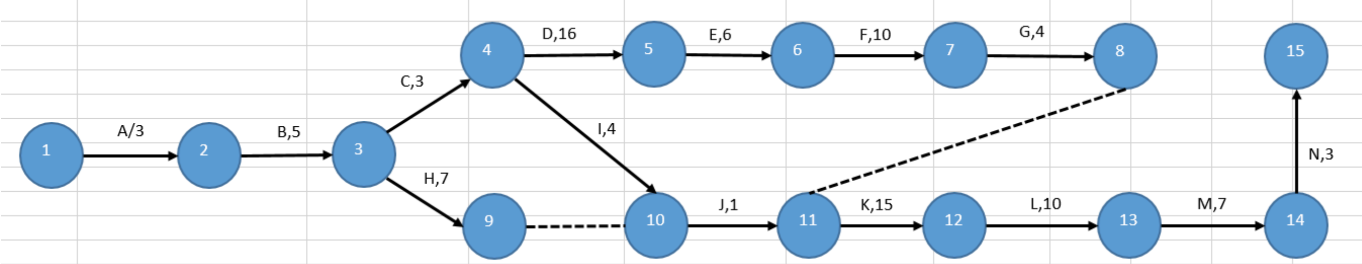
6.0 System Deployment and Evaluation

- 6.1 Deploy system
- 6.2 Collect feedback
- 6.3 Evaluate performance

7.0 Documentation

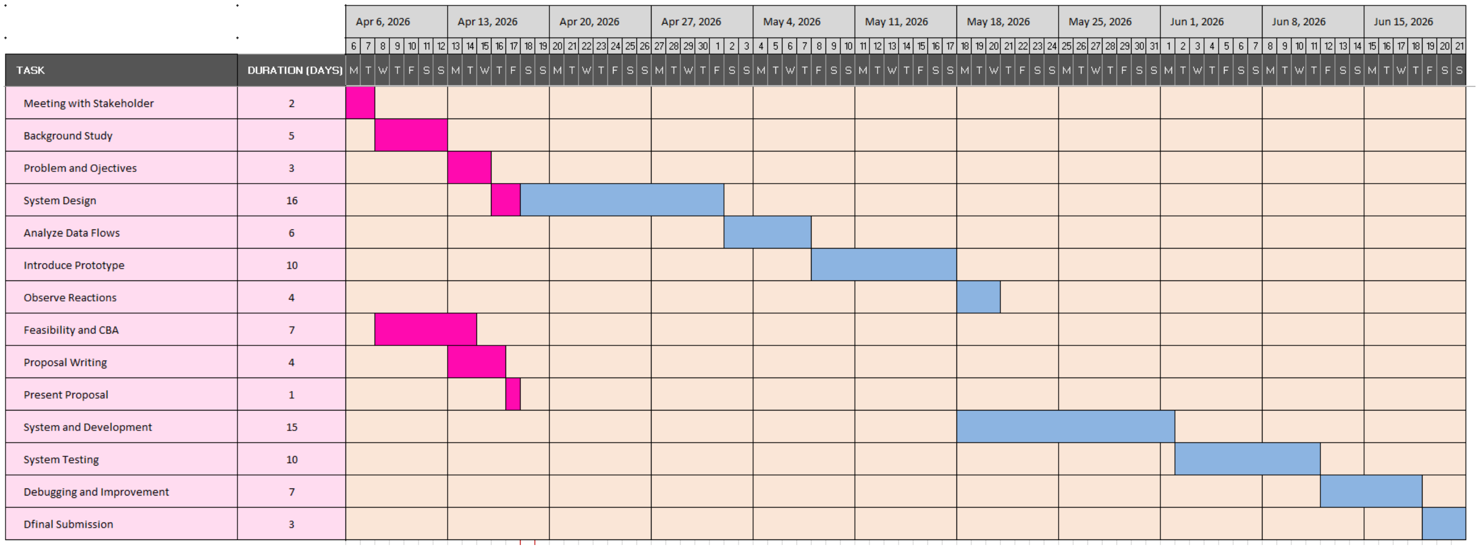
- 7.1 Prepare report

7.3 PERT CHART



ID	TASK	DURATION (DAYS)	DEPENDS ON
A	Meeting with Stakeholder	2	-
B	Background Study	5	A
C	Problem and Objectives	3	B
D	System Design	16	C
E	Analyze Data Flows	6	D
F	Introduce Prototype	10	E
G	Observe Reactions	4	F
H	Feasibility and CBA	7	B
I	Proposal Writing	4	C,H
J	Present Proposal	1	I
K	System Development	15	J
L	System Testing	10	K
M	Debugging and Improvement	7	L
N	Final Submission	3	M

7.4 GANTT CHART



8.0 BENEFITS & PROPOSED SYSTEM

The proposed FlexRent System provides a smarter and more modern solution to improve the current rental and booking process. Instead of relying on manual communication through applications like WhatsApp, the system centralizes all booking activities into one digital platform. This helps both customers and providers manage rental activities more efficiently while reducing common problems such as delayed responses, booking errors and miscommunication.

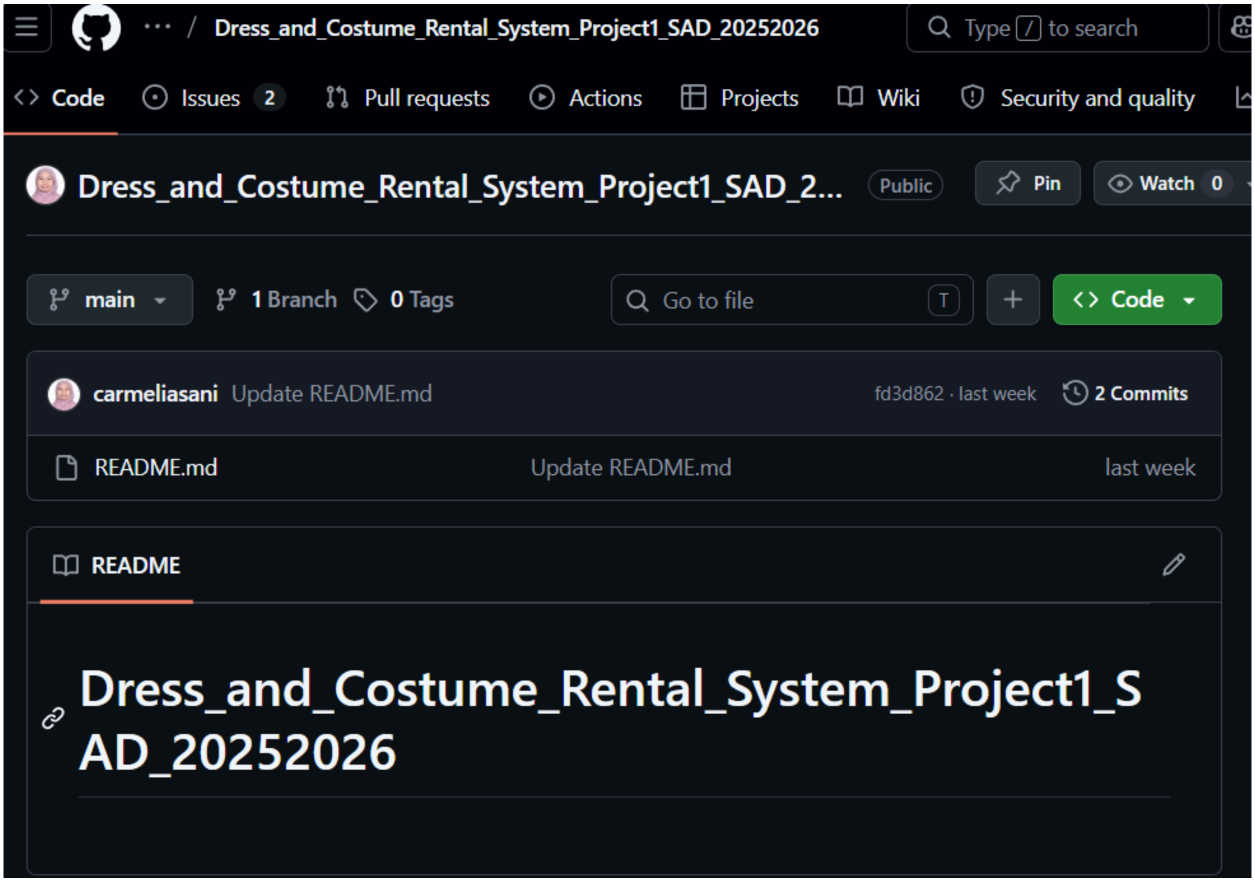
One of the key benefits of the system is the implementation of an automated booking and scheduling management feature. Through real time availability checking time slot management, users can instantly view available booking schedule without waiting for manual comfirmation from providers. This feature helps prevent overlapping booking while ensure smoother operation for providers. In addition, the inclusion of buffer time between bookings allows providers to prepare, clean or organize rental items properly before the next booking session, improving service quality and customer satisfaction.

Besides that, the system also includes a centralized inventory management feature that enables providers to monitor the status, availability and condition of rental items in real time. Providers can easily add, update or remove listings while tracking all booking activities through a single platform. This helps reduce confusion and improves overall business management. For customers, the system creates a smoother booking experience because they can browse available rental items, check availability and manage their bookings directly through the system without depending on manual communication.

Another important feature is the integration of Artificial Intelligence (AI) which are AI Smart Matching and AI Dynamic Pricing. AI Smart Matching recommends suitable dresses based on user preferences and booking history that helps customer find items more easily. Meanwhile, AI Smart Pricing automatically adjusts rental prices according to booking demand, seasonal trends and item availability, allowing providers to manage pricing more efficiently while ensuring fair prices for customers.

Furthermore the system supports secure online payment processing through an integrated payment gateway. Customers can make payment directly through the platform, while providers receive automatic payment verification without manually checking receipt or bank transfers. The system also support automated booking modifications, cancellations and refund calculations.

Overall, the proposed FlexRent System offers more efficient, user friendly and reliable solution compared to the current manual process. By combining automation, centralized management and AI features, the system helps improve operational efficiency, customer satisfaction and overall business performance.



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